

Name: [REDACTED]

Date: Feb 14, 2024

Total
28.5
/30

1. Evaluate the following limits.

$$\begin{aligned}
 \text{a. } \lim_{y \rightarrow 4} \frac{\sqrt{y+5}-3}{y^2-3y-4} \times \frac{\sqrt{y+5}+3}{\sqrt{y+5}+3} \\
 = \lim_{y \rightarrow 4} \frac{y+5-9}{(y-4)(y+1)(\sqrt{y+5}+3)} \\
 = \lim_{y \rightarrow 4} \frac{y-4}{(y-4)(y+1)(\sqrt{y+5}+3)} \\
 = \lim_{y \rightarrow 4} \frac{1}{(y+1)(\sqrt{y+5}+3)} \\
 = \frac{1}{(4+1)(\sqrt{4+5}+3)} \\
 = \frac{1}{30} //
 \end{aligned}$$

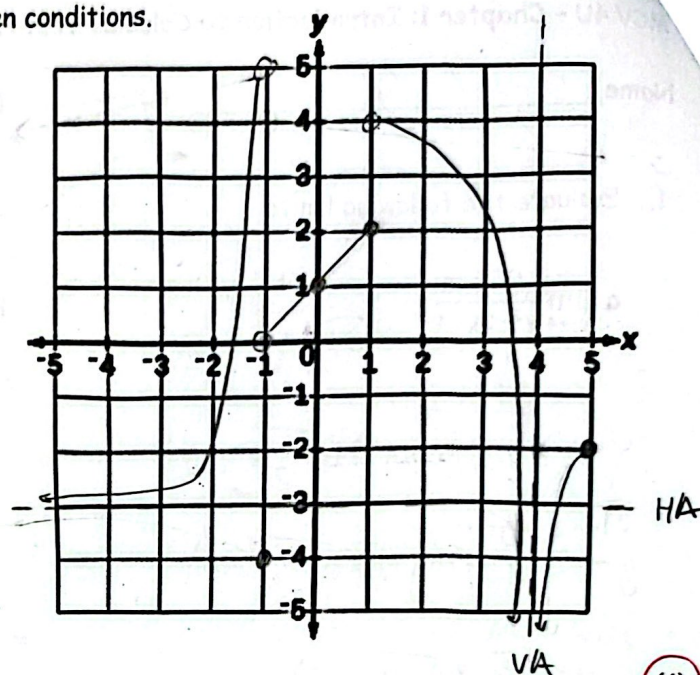
$$\begin{aligned}
 \text{b. } \lim_{x \rightarrow 1} \frac{x-\sqrt{x}}{\sqrt[3]{x}-1} \\
 = \lim_{x \rightarrow 1} \frac{x - x^{1/2}}{x^{1/3} - 1} \\
 \text{let } a = x^{1/6} \\
 \text{as } x \rightarrow 1, a \rightarrow 1 \\
 = \lim_{a \rightarrow 1} \frac{a^6 - a^3}{a^2 - 1} \\
 = \lim_{a \rightarrow 1} \frac{(a^3)(a^3-1)}{(a-1)(a+1)} \\
 = \lim_{a \rightarrow 1} \frac{(a^3)(a-1)(a^2+a+1)}{(a-1)(a+1)} \\
 = \lim_{a \rightarrow 1} \frac{a^3(a^2+a+1)}{a+1} \\
 = \frac{1^3(1^2+1+1)}{1+1} \\
 = \frac{3}{2} //
 \end{aligned}$$

(2) Determine the equation of the tangent to the curve $f(x) = \frac{2x-7}{3-x}$ where $x = -1$.

$$\begin{aligned}
 f(x) &= \frac{2x-7}{3-x} \\
 P(-1, -\frac{9}{4}) \quad Q(-1+h, \frac{2(h-1)-7}{3-(h-1)}) \\
 &= \lim_{h \rightarrow 0} \frac{\frac{2(h-1)-7}{3-(h-1)} - (-\frac{9}{4})}{\frac{h}{4(4-h)}} \\
 &= \lim_{h \rightarrow 0} \frac{\frac{2h-2-7}{3-h+1} + \frac{9}{4}}{\frac{h}{4(4-h)}} \\
 &= \lim_{h \rightarrow 0} \frac{(2h-9)4 + 9(4-h)}{4(4-h)} \times \frac{1}{h} \\
 &= \lim_{h \rightarrow 0} \frac{-36 + 8h - 9h + 36}{4(16-4h)} \times \frac{1}{h} \\
 &= \lim_{h \rightarrow 0} \frac{h(-1)}{4(16-4h)} \\
 &= \frac{-1}{16-4(0)} \\
 &= -\frac{1}{16} \leftarrow \text{slope} \\
 -\frac{9}{4} &= -\frac{1}{16}x + b \\
 b &= -\frac{35}{16} \\
 y &= -\frac{1}{16}x - \frac{35}{16} //
 \end{aligned}$$

3. Sketch the graph of any function that satisfies the given conditions.

- $f(-1) = -4, f(5) = -2$ ✓
- $\lim_{x \rightarrow -1^-} f(x) = 5$ ✓
- $\lim_{x \rightarrow 1^+} f(x) = 4$
- $\lim_{x \rightarrow 4^-} f(x) = -\infty, \lim_{x \rightarrow 4^+} f(x) = -\infty$ / VA
- $\lim_{x \rightarrow -\infty} f(x) = -3$ / HA
- When $-1 < x \leq 1, f(x) = x + 1$



4. Determine the value(s) of x for which the following function is discontinuous. Identify the type(s) of discontinuity. Justify your response.

$$f(x) = \frac{3x^2 + 17x - 6}{x^2 - 3x - 54}$$

$$f(x) = \frac{(3x-1)(x+6)}{(x-9)(x+6)}$$

hole @ $x = 6$

$$x - 9 = 0$$

$$x = 9 \quad \text{VA}$$

At $x = 6$ there is point discontinuity as there is a hole so $f(x)$ is not defined at $x = 6$, not meeting the requirements to be continuous. The other condition is that $\lim_{x \rightarrow 9} f(x) = f(9)$. At $x = 9$ there is a VA so there is infinite discontinuity. $\lim_{x \rightarrow 9} f(x) \neq f(9)$

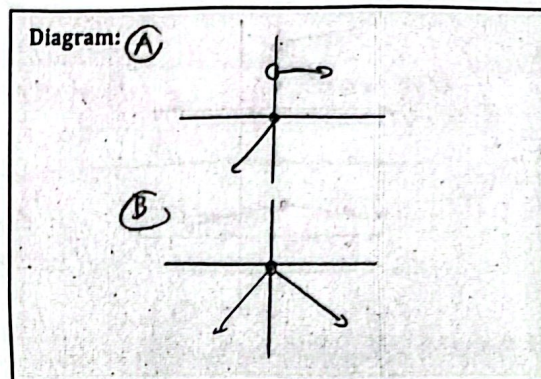
5. Does $\lim_{x \rightarrow 3} \begin{cases} 5, & \text{if } x = 3 \\ 2, & \text{if } x \neq 3 \end{cases}$ exist? Explain.

Yes, the limit does exist, even if $x = 3$ does not result in the same y value as $x \neq 3$. This is because limits are only concerned with what each side is approaching the x -value, not the actual value. This function stills value the $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$ as the $\lim_{x \rightarrow 3^-}$ and $\lim_{x \rightarrow 3^+}$ are the same (both 5) so the limit does exist.

bet 6. Determine if the statement below must be true, could be false or must be false. Support your conclusion using diagrams.

If $\lim_{x \rightarrow 0^-} f(x)$ exists and $\lim_{x \rightarrow 0^+} f(x)$ exists, then $\lim_{x \rightarrow 0} f(x)$ exists.

could be false. in order for $\lim_{x \rightarrow 0} f(x)$ to exist, $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$. If the LS and RS limits are diff (like (A)) then the $\lim_{x \rightarrow 0} f(x) = DNE$, if the limits are equal, the value of $\lim_{x \rightarrow 0} f(x)$ is also equal + exists. (B)



7. A dragster races down a 400 m strip in 8 seconds. Its distance in metres from the starting line after t seconds is $s(t) = 6t^2 + 2t$.

a) Calculate the average velocity when $1 \leq t \leq 4$.

$$AV = \frac{s(4) - s(1)}{4 - 1}$$

$$= \frac{[6(4)^2 + 2(4)] - [6(1)^2 + 2(1)]}{4 - 1}$$

$$= 32 \text{ meters / sec}$$

b) Determine when the velocity is 38 m/s.

$$P(x, 6(x)^2 + 2x) \quad Q(x+h, 6(x+h)^2 + 2(x+h))$$

$$= \lim_{h \rightarrow 0} \frac{6(x+h)^2 + 2(x+h) - [6(x)^2 + 2x]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{6(x^2 + 2xh + h^2) + 2x + 2h - 6x^2 - 2x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{6x^2 + 12xh + 6h^2 + 2h - 6x^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{12xh + 6h^2 + 2h}{h}$$

$$= \lim_{h \rightarrow 0} (12x + 6h + 2)$$

$$= 12x + 2$$

$$38 = 12x + 2$$

$$x = \frac{38 - 2}{12}$$

$$x = 3$$

$$x = 3 \text{ sec}$$

$\therefore$ the time when the $v = 38 \text{ m/s}$ is 3 s

8. Determine $\lim_{x \rightarrow 0} \frac{-\cos^2 x - 3\cos x + 4}{\sin^2 x}$

$$\sin^2 \theta + \cos^2 \theta = 1 \quad \tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\tan^2 \theta = \frac{\sin^2 \theta}{\cos^2 \theta} \times \cos^2 \theta \quad (2)$$

$$= -1 \left(\lim_{x \rightarrow 0} \cos(x) \right)^2 - 3 \left(\lim_{x \rightarrow 0} \cos(x) \right) + \left(\lim_{x \rightarrow 0} 4 \right)$$

$$\left(\lim_{x \rightarrow 0} \sin^2 \theta \right) \times \left(\lim_{x \rightarrow 0} \cos^2 \theta \right)$$

but

$$= -1(1)^2 - 3(1) + 4$$

$$\frac{0}{1} \times 1$$

ON SCRAP PAPER

= DNE

9. Given the function $f(x) = \frac{x^3 + ax^2 + bx + 5}{x^3 + 2x^2 - x - 2}$, determine whether there are numbers a and b such that both $\lim_{x \rightarrow 1} f(x)$ and $\lim_{x \rightarrow -1} f(x)$ exists. If so, evaluate both limits.

$$f(x) = \frac{x^3 + ax^2 + bx + 5}{x^3 + 2x^2 - x - 2}$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x) \quad \lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^+} f(x)$$

$$f(1) = \frac{1 + a + b + 5}{1 + 2(1) - 1 - 2}$$

$$= \frac{b + a + b}{0}$$

$$f(-1) = \frac{(-1)^3 + a(-1)^2 + b(-1) + 5}{(-1)^3 + 2(-1) - (-1) - 2}$$

$$= \frac{-1 + a - b + 5}{-1 - 2 + 1 - 2}$$

$$= \frac{a + b + 4}{0}$$

$$b + a + b = a - b + 4$$

$$\frac{2b}{2} = \frac{-2}{2}$$

$$\boxed{b = -1}$$

$$\boxed{a = -5}$$

idk sorry :-

$$\lim_{x \rightarrow -1} = \frac{(-1)^3 + (-5)(-1)^2 + (-1)(-1) + 5}{(-1)^3 + 2(-1) - (-1) + 2}$$

$$= 0//$$

$$\lim_{x \rightarrow 1} = \frac{(1)^3 + (-5)(1)^2 - 1(1) + 5}{(1)^3 + 2(1) - (1) + 2}$$

$$= 0//$$

the limit of both functions = 0

⑤ let a rep $\cos x$ as $x \rightarrow 0$

$\cos x \rightarrow 1$

$$= \lim_{a \rightarrow 1} \frac{a^2 - 3a + 4}{1 - a^2}$$

$$= \lim_{a \rightarrow 1} \frac{a^2 + 3a - 4}{(a-1)(a+1)}$$

$$= \lim_{a \rightarrow 1} \frac{(a-1)(a+4)}{(a-1)(a+1)}$$

$$= \lim_{a \rightarrow 1} \frac{1+4}{1+1}$$

$$= \lim_{a \rightarrow 1} \frac{5}{2} \quad \checkmark$$

$$= 5/2 //$$